

Functional Analysis — Homework 3

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Problem 1. Let X, Y be normed spaces and $\mathcal{B}(X, Y)$ the vector space of all bounded linear operators $T : X \rightarrow Y$. Show that the operator norm

$$\|T\|_{op} := \sup_{\|x\| \leq 1} \|T(x)\|$$

is a norm in $\mathcal{B}(X, Y)$. Show that if Y is Banach, then $\mathcal{B}(X, Y)$ is Banach.

Proof. We have

$$\begin{aligned} \|cT\|_{op} &= \sup_{\|x\| \leq 1} \|cT(x)\| \\ &= \sup_{\|x\| \leq 1} |c| \cdot \|T(x)\| \\ &= |c| \sup_{\|x\| \leq 1} \|T(x)\| \\ &= |c| \cdot \|T\|_{op} \end{aligned}$$

and

$$\begin{aligned} \|T + S\|_{op} &= \sup_{\|x\| \leq 1} \|T(x) + S(x)\| \\ &\leq \sup_{\|x\| \leq 1} \|T(x)\| + \|S(x)\| \\ &\leq \sup_{\|x\| \leq 1} \|T(x)\| + \sup_{\|x\| \leq 1} \|S(x)\| \\ &= \|T\|_{op} + \|S\|_{op}. \end{aligned}$$

Now assume Y is Banach and let $(f_n) \subset \mathcal{B}(X, Y)$ be Cauchy. We want to show that (f_n) has a pointwise limit f . First, we have $(f_n(0)) \rightarrow 0$ since $f_n(0) = 0$ for all n , so we are justified in setting $f(0) = 0$.

Now given $\varepsilon > 0$, we can choose $M \in \mathbb{Z}^+$ such that when $m, n > M$,

$$\begin{aligned} \frac{\varepsilon}{\|x\|} &> \|f_m - f_n\|_{op} \\ &= \sup_{x \neq 0} \frac{\|f_m(x) - f_n(x)\|}{\|x\|} \\ &\geq \frac{\|f_m(x) - f_n(x)\|}{\|x\|} \\ \implies \varepsilon &> \|f_m(x) - f_n(x)\|, \end{aligned}$$

where the third and fourth lines apply for any *particular* nonzero x , so we have that $(f_n(x))$ is Cauchy and thus converges, so we can set $f(x) = \lim_{n \rightarrow \infty} f_n(x)$.

We have thus demonstrated that (f_n) converges to a pointwise limit, so since (f_n) was an arbitrary Cauchy sequence, we have that $\mathcal{B}(X, Y)$ is Banach. $\square$

Problem 2. Let X, Y be normed spaces with X infinite-dimensional and $Y \neq \{0\}$. Using the axiom of choice (or equivalent statement), show that there exists a linear operator $T : X \rightarrow Y$ which is unbounded.

Proof. Let B be a normalized Hamel basis for X (requires AoC) and choose a countable subset $C = \{x_1, x_2, x_3, \dots\} \subset B$. Also let $y \in Y$ such that $\|y\| = 1$. Then there is a linear map $T : X \rightarrow Y$ determined by

$$\begin{aligned} T(x_n) &= ny & \forall x_n \in C \\ T(x) &= 0 & \forall x \in B \setminus C, \end{aligned}$$

for which we note that for $x_n \in C$,

$$\|T\|_{op} \geq \|T(x_n)\| = \|ny\| = |n| \cdot \|y\| = n.$$

Since this is true for every n , T is unbounded. $\square$

Problem 3. Let $C^\infty[0, 1]$ be the vector space of all infinitely differentiable functions from $[0, 1]$ to $\mathbb{R}$. Consider the derivative as a linear operator

$$\mathcal{D} : C^\infty[0, 1] \rightarrow C^\infty[0, 1].$$

Show that it is impossible to equip $C^\infty[0, 1]$ with a norm $\|\cdot\|$ such that $\mathcal{D}$ is bounded.

Proof. Given $N \in \mathbb{Z}^+$, let $f(x) = e^{Nx}$. Then

$$\begin{aligned} \sup_{g \neq 0} \frac{\|\mathcal{D}(g)\|}{\|g\|} &\geq \frac{\|\mathcal{D}(f)\|}{\|f\|} \\ &= \frac{\|Ne^{Nx}\|}{\|e^{Nx}\|} \\ &= \frac{|N| \cdot \|e^{Nx}\|}{\|e^{Nx}\|} \\ &= |N| \\ &= N. \end{aligned}$$

Since this is the case for any N , $\mathcal{D}$ is unbounded. $\square$

Problem 4. Recall that c_{00} is dense in ℓ^1 . Consider both spaces equipped with the ℓ^1 norm. Show that the identity map

$$\begin{aligned} \iota : c_{00} &\rightarrow c_{00} \\ x &\mapsto x \end{aligned}$$

cannot be extended to a continuous map $\tilde{\iota} : \ell^1 \rightarrow c_{00}$.

Proof. Assume that we could extend ι to a continuous map $\tilde{\iota} : \ell^1 \rightarrow c_{00}$. Then given $x \in \ell^1 \setminus c_{00}$, we have $x = \lim_{n \rightarrow \infty} (x_i)_n$, where each sequence $(x_i)_n \in c_{00}$, so

$$\begin{aligned}\tilde{\iota}(x) &= \tilde{\iota}\left(\lim_{n \rightarrow \infty} (x_i)_n\right) \\ &= \lim_{n \rightarrow \infty} \tilde{\iota}((x_i)_n) \\ &= \lim_{n \rightarrow \infty} \iota((x_i)_n) \\ &= \lim_{n \rightarrow \infty} (x_i)_n \\ &= x \notin c_{00},\end{aligned}$$

where the second equality follows from continuity/sequential continuity. Since x was chosen to not be in c_{00} , $\tilde{\iota}$ is not actually a map into c_{00} , so no such map exists.

Generalization: This argument appears to demonstrate that we cannot extend the identity map $\iota : X \rightarrow X$ to a sequentially continuous map $\tilde{\iota} : Y \rightarrow X$ for *any* choice of X and Y where X is dense in Y . $\square$

Problem 5. Let X be a Banach space and $T : X \rightarrow \ell^\infty$ a linear operator. Writing

$$T(x) = (f_1(x), f_2(x), f_3(x), \dots) \in \ell^\infty,$$

we get a sequence of linear functionals $f_n : X \rightarrow \mathbb{R}$. Show that T is bounded if and only if each f_n is a bounded linear functional on X .

Proof. If T is bounded, then for any $x \in X$, we have $\|T(x)\| = \sup_{n \in \mathbb{Z}^+} |f_n(x)|$, so $|f_n(x)| \leq \|T(x)\|$ for any particular n . This tells us that for any n ,

$$\sup_{x \neq 0} \frac{|f_n(x)|}{\|x\|} \leq \sup_{x \neq 0} \frac{\|T(x)\|}{\|x\|} < \infty,$$

where the first inequality follows from term-by-term comparison, and the second inequality is the fact that T is bounded. Thus f_n is bounded for all n .

Now assume that f_n is bounded for all n . Then

$$\begin{aligned}\sup_{\|x\|=1} \|T(x)\| &= \sup_{\|x\|=1} \left(\sup_{n \in \mathbb{Z}^+} |f_n(x)| \right) \\ &\leq \sup_{\|x\|=1} \left(\sup_{n \in \mathbb{Z}^+} \|f_n\|_{op} \|x\| \right) \\ &= \sup_{n \in \mathbb{Z}^+} \|f_n\|_{op} \dots\end{aligned}$$

Not sure how to get the inequality I need here! $\square$